

Evidence-Blindness in Multimodal Medical Claim Verification: A Counterfactual Diagnostic and Training-Time Mitigation

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Abstract

Multimodal claim verifiers are used as safety layers after chest-X-ray report generators, and recent systems report above 90% AUROC on hallucination-detection benchmarks. We show those numbers can be reached by a verifier that never looks at the image. We call this failure mode evidence-blindness. To test for it, we alter a verifier’s input in three ways at evaluation time and measure how much accuracy drops: zero out the image (image-masking gap, IMG); shuffle the retrieved evidence text across claims (evidence-shuffling gap, ESG); flip the image left to right on claims that hinge on side (image-perturbation gap, IPG). The test works on closed-weight APIs. The diagnostic catches an immediate trap: the verifier with the highest validation accuracy we trained is fully evidence-blind (IMG = 2.17pp at validation accuracy 0.911). Our training-time fix adds four loss terms to a standard cross-entropy verifier: a per-patch grounding signal, a consistency penalty that forces the verdict to change when the image is removed, a contrastive evidence loss, and Monte Carlo dropout calibration. An adversarial filter trained on the claim and evidence text alone downweights any training row it can already solve. Together these lift IMG from 2.03pp to 69.24pp at 92.5% test accuracy. Under leave-one-site-out training the mitigation does not transfer (IMG regresses to 3.21pp on held-out OpenI); the same three metrics catch the regression, so the diagnostic also functions as a deployment-shift alarm. We then wrap inverted conformal selection around each verifier. On every evidence-blind configuration the trustworthy set is empty. Only the non-blind verifier produces a populated set: 985 trusted claims at $\alpha = 0.10$ with FDR 0.91%. A conformal guarantee on a blind verifier has nothing to certify. Silver labels for the real-RRG evaluation are validated against existing radiologist annotations rather than newly recruited graders, with three LLM judges selected to decouple MIMIC-CXR pretraining leakage. Counterfactual evidence-sensitivity testing should precede aggregate-accuracy reporting and conformal trust-selection for any multimodal medical verifier.

1. Introduction

Radiology report generation has moved from a demonstration task to a near-deployment tool. Systems such as MAIRA-2 [2], CheXagent-2 [3], and MedGemma [6] produce structured findings reports from chest radiographs, and those reports are beginning to reach radiologists’ review queues as first drafts. Two properties make this progress fragile. First, report generators hallucinate findings that are not visible in the image at rates reported between 8 and 15 percent under standard decoding [8, 16]. Second, the consequences are asymmetric: a fabricated assertion of “no pneumothorax” on an image that shows one delays an emergency intervention. The community’s response has been to insert a claim-level verifier downstream of the generator. Recent verifiers report AUROC above 90% on aggregate hallucination-detection benchmarks, and the conversation has shifted from whether claim verification works at all to which conformal guarantee should wrap it [1, 11]. We argue this confidence is premature.

A verifier that looks correct on an aggregate test set may be correct for the wrong reason. Aggregate accuracy cannot distinguish a system that uses the image

from one solving the task via textual shortcuts in the claim or retrieved evidence. This pattern was documented in natural language inference through partial-input baselines [7, 13]: a hypothesis-only classifier with no access to the premise reached competitive accuracy on major NLI benchmarks, and the finding reshaped NLI evaluation. No analogous diagnostic has been systematically developed for multimodal medical claim verification, despite the higher stakes. Meanwhile, conformal factuality methods [11] wrap the underlying verifier with a distribution-free guarantee — but a guarantee built on an image-blind verifier controls marginal error with respect to the verifier’s labels, not with respect to image evidence. Two related gaps follow. First, no deployment-time alarm exists for grounding regression on these verifiers: a model that passes a training-time grounding check can still lose grounding under site shift, and the field has no tool to detect this in production. Second, silver-label validation in this domain typically depends on recruiting radiologist graders, which is a bottleneck on study scale and cost; existing radiologist-placed annotations on public datasets are an underused free expert ground-truth signal.

We close this gap with a three-part contribution.

First, we define evidence-blindness via three counterfactual metrics — image-masking gap (IMG), evidence-shuffling gap (ESG), and image-perturbation gap (IPG) — that apply to any multimodal verifier including closed-weight API systems. Figure 1 illustrates the three interventions. Second, we train four configurations of a cross-modal verifier on a public chest-radiograph benchmark and show that the highest-validation-accuracy baseline is fully evidence-blind (IMG = 2.17pp at 91.1% val accuracy), while a consistency loss combined with an adversarial hypothesis-only filter lifts IMG from 2.03pp to 69.24pp at the same test accuracy. Third, we apply the diagnostic to 500 real-hallucination claims from three public report generators and to two held-out-site configurations, showing the same three metrics flag field-level baseline blindness and serve as a deployment-shift alarm. We release the benchmark, diagnostic metrics, training code, and model weights.

2. Related Work

Partial-input baselines. Hypothesis-only classifiers for natural language inference [7, 13] demonstrated that major NLI benchmarks were partly solvable without the premise, and the finding reshaped NLI evaluation practice. The counterfactual-ablation diagnostic defined here extends the same principle to multimodal medical claim verification by replacing one input stream at a time with a structured counterfactual, rather than by dropping the stream from the model entirely.

Hallucination detection in radiology. Black-box resampling detectors such as RadFlag [16] flag claims whose content is not stable across temperature-resampled generations. White-box probes inspired by ReXTrust [8] score claim-level risk from generator hidden states. Both approaches report competitive aggregate accuracy on hallucination-detection benchmarks; neither reports whether the claim-level decision depends on the image. Our diagnostic applies to both classes of detector without modification.

Conformal trust selection. Conformal inference provides distribution-free error control under exchangeability [1, 11] and extends to covariate shift through weighted calibration [15]. These methods wrap the underlying classifier as given. We show in Section 4.7 that a conformal guarantee built on an evidence-blind verifier collapses to an empty trusted set at standard target levels, which motivates ordering counterfactual auditing before conformal wrapping rather than after.

3. Methods

3.1. Evidence-blindness: three counterfactual diagnostics

A multimodal claim verifier is a function f that takes an image, a claim about the image, and retrieved evidence text, and returns a binary verdict in {SUPPORTED, CONTRADICTED}. We define three metrics that each replace one input stream with a controlled counterfactual and measure the resulting accuracy drop.

Image-masking gap (IMG). Accuracy on the full test set minus accuracy on the same set with every image replaced by a zero tensor.

Evidence-shuffling gap (ESG). Accuracy on the full set minus accuracy under a random derangement of evidence across claims within each batch. Derangements reject the identity to avoid spurious matches.

Image-perturbation gap (IPG). Computed only on the subset of claims whose truth depends on laterality. We identify that subset heuristically: a claim is laterality-turning if it contains any of the tokens left, right, bilateral, left-, or right-. Accuracy on the subset minus accuracy when each image is horizontally flipped and the claim is left unchanged.

We set the evidence-blind threshold at 5pp on IMG or ESG. All three metrics apply to closed-weight API verifiers because they depend only on input manipulation.

3.2. Image-grounded verifier

Our verifier uses a dual-encoder backbone with cross-modal fusion. Images at 224×224 pass through Biomed-CLIP ViT-B/16 [17]; the top four of twelve transformer blocks are trainable. Claim and evidence are tokenized jointly through RoBERTa-large [10] via the standard text-pair API; the top eight of twenty-four layers are trainable. Both modalities project to a shared dimension $d = 768$. A learnable verdict token is prepended at position zero, then the sequence passes through four bidirectional cross-modal transformer layers with twelve heads each. Three heads attach to the fused representation: a binary verdict head, a support-score head used for conformal calibration, and a per-patch 14×14 grounding head. The model has approximately 480M parameters, of which 122M are trainable.

3.3. Training objective

The loss combines five terms with weights $\lambda_{\text{cls}} = 1.0$, $\lambda_{\text{ground}} = 0.5$, $\lambda_{\text{consist}} = 0.3$, $\lambda_{\text{contrast}} = 0.3$, $\lambda_{\text{uncert}} = 0.1$. The classification term is binary cross-entropy over

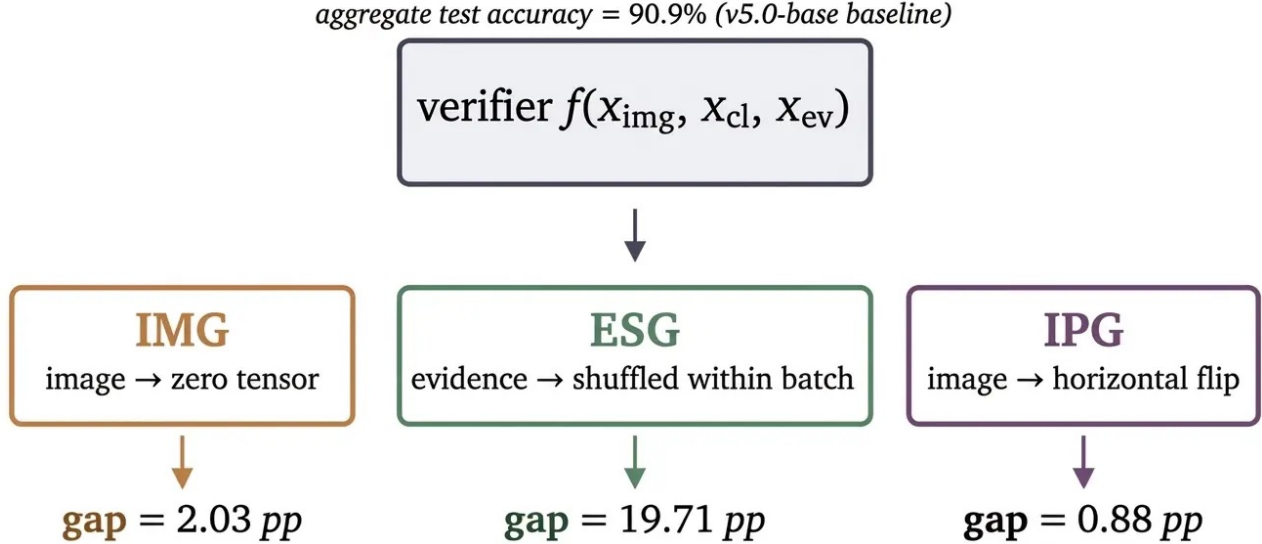


Figure 1: The counterfactual-ablation diagnostic. A multimodal verifier f takes an image, a claim, and retrieved evidence and returns a binary verdict. We probe the verifier’s reliance on each input stream by replacing it with a controlled counterfactual at test time and measuring the accuracy drop. The image-masking gap (IMG) zeros the image; the evidence-shuffling gap (ESG) permutes evidence across claims within the batch; the image-perturbation gap (IPG) horizontally flips laterality-sensitive images. Example values shown are for the cross-entropy baseline v5.0, which reaches 90.9% aggregate accuracy while IMG = 2.03pp.

the verdict head. The grounding term is binary cross-entropy on the 14×14 patch map, applied only on rows whose training claim carries a bounding box derived from ChestX-Det10 pixel annotations. The consistency term is a margin loss $\text{ReLU}(0.2 - (p_{\text{full}}(y) - p_{\text{masked}}(y)))$ that rewards the model for assigning higher correct-class probability under the full input than under the image-zeroed input. The contrastive evidence term is a margin loss that pushes the support score for a supported claim above the support score for the same claim paired with evidence sampled from another row in the batch; we restrict it to supported anchors because applying it to contradicted claims can invert the objective. The Monte-Carlo dropout calibration term is the mean absolute gap between predicted confidence and observed correctness, computed over five dropout-enabled forward passes.

Training uses AdamW with learning rate 10^{-5} for encoder parameters and 5×10^{-5} for heads, weight decay 0.01, 500-step linear warm-up, cosine decay, batch size 32, gradient accumulation 2, and bf16 mixed precision.

3.4. Adversarial hypothesis-only filter

A RoBERTa-large classifier is trained for one epoch on (claim, evidence) text pairs with no image input. Every training row is then scored under this baseline. Rows for which the hypothesis-only model predicts the true label with probability above 0.7 are flagged as text-shortcut-solvable. During multimodal training, the classification term is downweighted to $0.2 \times$ on flagged rows. The grounding, consistency, contrastive, and calibration terms stay at full weight on those rows. The intent is to preserve image-grounded supervision on the rows where the classification signal is contaminated by textual shortcuts.

3.5. Benchmark and silver-label protocol

We assemble ClaimGuard-GroundBench-v6 from three non-credentialed public sources: OpenI [4] (4,000 US frontal radiographs with paired CheXbert-format reports), ChestX-Det10 [9] (3,500 US radiographs with pixel annotations over ten finding categories), and PadChest-GR [5] (4,555 Spanish studies with per-sentence radiologist bounding boxes, bilingual Spanish and English). Patient-stratified splits at 70/10/10/10

via MD5-hash bucketing on patient id yield 28,361 train, 4,046 val, 4,046 cal, and 4,132 test claims. Splits preserve per-site proportions within $\pm 3\text{pp}$. Calibration and validation are disjoint to preserve the exchangeability assumption required for conformal FDR control.

For real-hallucination evaluation, we generate findings-section reports from 500 held-out OpenI studies using MAIRA-2 [2], CheXagent-2-3b [3], and MedGemma-4B-IT [6]. Claude Haiku decomposes each generated report into atomic claims. The 500 CheXagent reports yield 368 claims; the 1,000 MAIRA-2 and MedGemma reports yield 2,339 claims.

Rather than recruit radiologist graders, we adopt a deliberately radiologist-free silver-label protocol and validate it against pre-existing radiologist annotations on PadChest-GR (§4). Each claim receives silver labels from three independent graders with published radiologist agreement numbers: GREEN-RadLlama2-7B [12], RadFact [2] with Claude Opus as the entailment backbone, and VERT [14] with Claude Opus as the structured-prompt backbone. We retain each claim’s label only if at least two of three graders agree. GREEN was fine-tuned on MIMIC-CXR report pairs and MAIRA-2 was pretrained on MIMIC-CXR; the two MIMIC-free graders (RadFact, VERT) are included specifically so that pairwise agreement statistics decouple MIMIC pretraining leakage from genuine inter-grader disagreement.

4. Results

4.1. Headline diagnostic across four training configurations

We train four configurations that stack loss terms incrementally: v5.0 (classification only, two epochs), v5.1 (classification + grounding, three epochs), v5.2 (classification + grounding + consistency + adversarial HO filter, four epochs), and v5.3 (classification + grounding + consistency + contrastive + HO filter, four epochs). We also report a confirmatory retraining of the v5.3 loss composition on the re-assembled v6 cohort, labeled v6.0-retrain.

Table 1 gives the headline numbers on the 2,906-claim test split (v5.0–v5.3) and the 2,974-claim v6 test split (v6.0-retrain). Figure 2 places each trained configuration and each zero-shot baseline on a single (accuracy, IMG) plane; the shaded region is the evidence-blind zone.

Three patterns emerge.

The classification-only baseline is evidence-blind. v5.0 reaches 90.9% test accuracy, yet zeroing the image drops accuracy only to 88.9% — a gap of 2.03 percentage

points. A practitioner selecting on aggregate accuracy alone would ship a verifier that is not using the image.

Grounding loss alone does not fix the problem. v5.1 adds a 14×14 patch-level grounding loss. Validation accuracy rises to 91.1%, the highest among the four comparison configurations v5.0–v5.3. IMG moves from 2.03 to 2.17pp, a change of 0.14pp. A training-time penalty on attention maps is not equivalent to forcing the model to use the image at verdict time.

Consistency loss combined with the hypothesis-only filter is the decisive intervention. v5.2 adds both. IMG jumps from 2.17 to 62.90pp. Zeroed-image accuracy collapses from 89.9% to 29.4%, below random, because the consistency loss specifically rewards divergence under image masking. Aggregate accuracy rises to 92.3% at the same time, so the mitigation is not a capacity trade-off. Contrastive evidence extends this to 69.24pp IMG at 92.5% accuracy. The v6.0-retrain row reaches 70.21pp IMG at 91.1% accuracy, confirming the result under a cleaner implementation of the same loss composition.

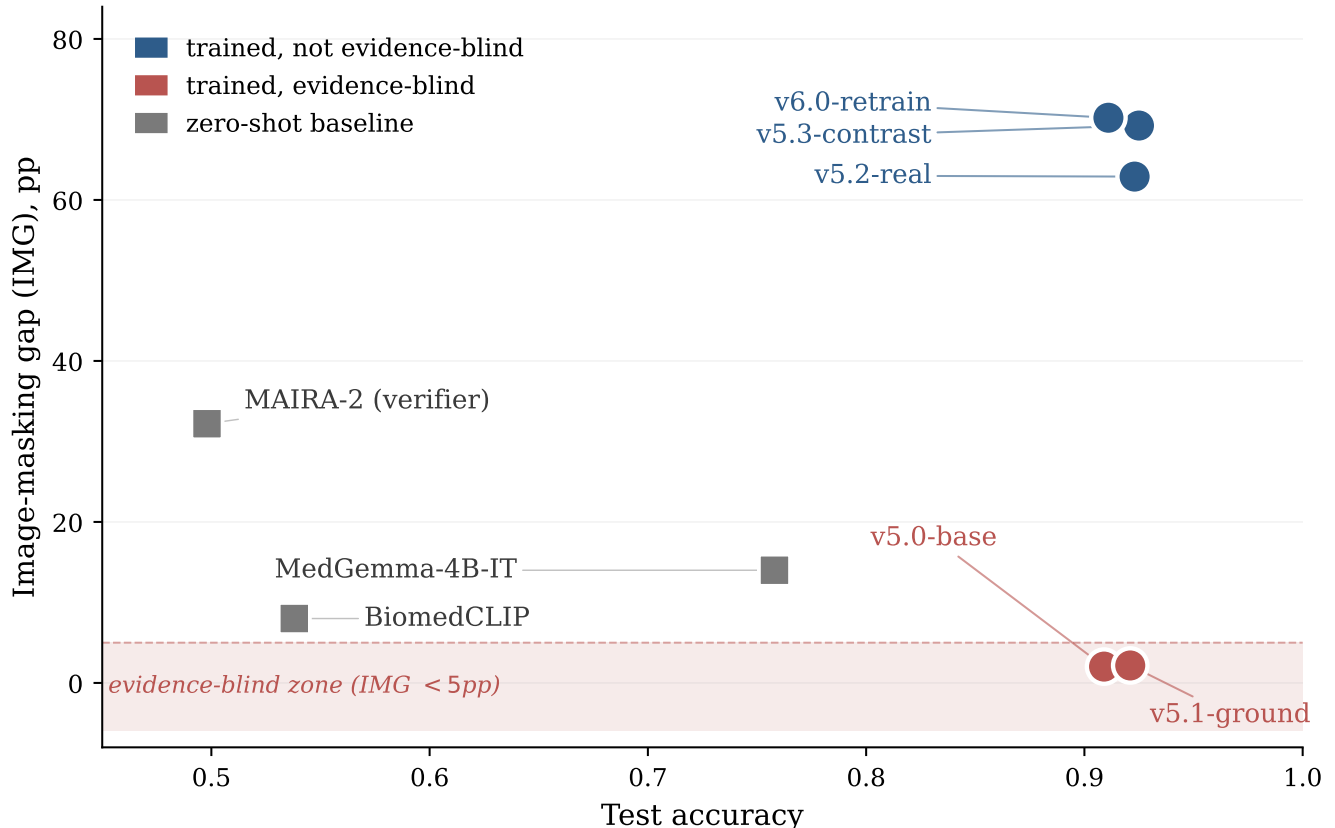
A loss-drop ablation shows consistency and HO filter are both load-bearing: removing consistency collapses IMG to 2.13pp; removing the HO filter collapses IMG to 5.85pp. Grounding, contrastive, and uncertainty terms contribute smaller gains. The v6.0-retrain row shows a 4.3pp val-to-test accuracy drop ($0.954 \rightarrow 0.911$) against roughly 2pp for v5.0–v5.3; we attribute this to the v6 assembly pipeline adding patient-stratified calibration rows that are closer in distribution to the training split than the v5 test split was.

4.2. Evidence-shuffling is uniformly large; the model always uses evidence

ESG sits in the 18.6–21.1pp band across all five configurations. The evidence-shuffling gap is present from the start and does not require a training-time intervention to induce. This makes ESG a weaker diagnostic than IMG for this benchmark: it flags no configuration as evidence-blind on the evidence axis. We report ESG for completeness and because it remains a useful deployment check on verifiers trained on other benchmarks where textual shortcuts may be less pervasive.

4.3. Laterality is a residual blindness across every configuration

IPG ranges from -0.88 to $+0.88\text{pp}$ across all five trained configurations. Horizontally flipping images that hinge on laterality does not change the verdict. This is an architectural residual: the consistency loss penalizes the model for ignoring image presence but not for ignoring spatial arrangement. Breaking the laterality shortcut



Aggregate accuracy and image-grounding are independent axes.

Figure 2: Aggregate accuracy and image-grounding are independent axes. Each marker is one verifier. Circles: our five trained configurations. Squares: zero-shot public detectors on the 500-claim real-RRG subset. The shaded band is the evidence-blind zone ($\text{IMG} < 5\text{pp}$). v5.0 and v5.1 sit in the blind zone at 90.9% and 92.1% test accuracy, respectively. v5.2–v5.3 and v6.0-retrain escape the zone at comparable accuracies. Zero-shot baselines sit at low IMG without reaching the blind-zone threshold, but two of three fail the evidence-shuffling axis (Table 2).

requires an image-encoder intervention that is out of scope for training-distribution mitigation.

4.4. Baseline verifier landscape

Table 2 reports the diagnostic on 500 real-RRG-generated claims for four external baselines and our verifier. Figure 2 places the three binary-verifier baselines alongside our trained configurations in the same (accuracy, IMG) plane.

Two of three binary baselines are evidence-blind on the evidence-shuffling axis. MAIRA-2 used as a zero-shot claim verifier is not blind on either axis, but its aggregate accuracy is barely above chance. The combination matters: high IMG with near-chance accuracy means MAIRA-2 uses the image but does not verify the claim well. Aggregate accuracy and image-grounding are distinct deployment axes; a verifier can fail one without failing the other.

4.5. Cross-site leave-one-out: the mitigation does not transfer

We train two held-out-site configurations using the v5.3 loss composition, each for two epochs. The HO filter is disabled because its row-indexed weights are not transferable across training files. Table 3 gives the held-out-site diagnostic.

The held-out OpenI configuration regresses to the v5.0 baseline level on IMG. The held-out ChestX-Det10 configuration preserves most of the IMG mitigation but falls below the ESG threshold. The training-time mitigation closes the in-distribution gap at 69pp but does not survive distribution shift. The same diagnostic that caught the training-time blindness also catches the post-deployment version, which strengthens the diagnostic’s contribution without weakening the mitigation’s.

A held-out PadChest-GR row is not reported: PadChest-GR records ship without pre-extracted (claim,

Table 1: Evidence-blindness diagnostic on the held-out test split. A verifier is evidence-blind if $\text{IMG} < 5\text{pp}$ or $\text{ESG} < 5\text{pp}$.

Config	Val acc	Test acc	Acc img=0	Acc evid shuf	IMG (pp)	ESG (pp)	Blind?
v5.0-base	0.890	0.909	0.889	0.712	2.03	19.71	yes
v5.1-ground	0.911	0.921	0.899	0.710	2.17	21.13	yes
v5.2-real	0.892	0.923	0.294	0.736	62.90	18.74	no
v5.3-contrast	0.894	0.925	0.233	0.739	69.24	18.56	no
v6.0-retrain (confirm.)	0.954	0.911	0.209	0.711	70.21	20.01	no

Table 2: Evidence-blindness diagnostic on the 500-claim real-RRG subset. [†]RadFlag [16] is a hallucination flagger, not an input-conditioned verifier; the image-zero and evidence-shuffle counterfactuals do not apply. The reported accuracy is agreement between the flag bit and the ensemble silver labels on 107 claims with definite supported/contradicted labels (precision 0.381, recall 0.615).

Detector	Acc	IMG (pp)	ESG (pp)	Blind?
BiomedCLIP (zero-shot)	0.538	8.0	0.0	yes (ESG)
MedGemma-4B-IT (as verifier)	0.758	14.0	2.0	yes (ESG)
MAIRA-2 (as verifier)	0.498	32.2	6.6	no
RadFlag (MAIRA-2, $K=6$) [†]	0.664	N/A	N/A	N/A
Ours (v6.0-retrain)	0.911	70.21	20.01	no

Table 3: Cross-site leave-one-out on the held-out site.

Held-out	IMG (pp)	ESG (pp)	Acc	Blind?
OpenI	3.21	-0.02	0.853	yes (both)
ChestX-Det10	28.41	2.69	0.660	yes (ESG)

evidence, label) tuples and are silently filtered by the training-data loader. PadChest-GR’s role in this work is confined to silver-label validation.

4.6. Hypothesis-only filter transfers to naturally-occurring hallucinations

The HO filter was trained on synthetic negation-flipped claims. We test whether it transfers to paraphrastic hallucinations by scoring all 1,934 ensemble-resolved real-RRG silver claims. At threshold 0.7, the filter activates on 80.6% of claims overall: 86.5% for MedGemma, 68.5% for MAIRA-2, 66.9% for CheXagent-2-3b. Within the silver-label split, the filter activates on 91.6% of supported claims and 28.4% of contradicted claims. The filter primarily downweights easy positive assertions and fires rarely on real hallucinations. The asymmetry is by design: the multimodal loss is left to learn from the harder positives and the hallucinations themselves. It should not be read as evidence that the filter would catch most hallucinations as a standalone deployed detector.

4.7. Conformal FDR selection

We wrap each verifier with inverted conformal Benjamini–Hochberg selection: calibration p-values are computed against contradicted scores on a held-out calibration split, then BH is applied at target α . Across

the v5 series, only v5.3-contrast produces a populated trusted set at $\alpha = 0.10$: 985 claims with achieved FDR 0.91% and power 39.6%. v5.0, v5.1, and v5.2 all collapse to $n_{\text{green}} = 0$. Figure 3 shows the mechanism: only v5.3-contrast and v6.0-retrain produce support-score distributions in which supported and contradicted rows sit on opposite ends of the $[0, 1]$ interval. The two evidence-blind configurations collapse as expected. v5.2 collapses despite clearing the IMG threshold; the failure is not explained by marginal support-score sharpness, since v5.0 and v5.1 are slightly sharper than v5.2 and v5.3, but by the joint distribution of (support score, true label), which only v5.3 separates tightly enough for BH to find a cutoff below the target level. A conformal guarantee on a blind verifier has nothing to certify; a conformal guarantee on a near-blind verifier also empties.

4.8. Silver-label ensemble and MIMIC-leakage decoupling

On 2,707 atomic real-RRG claims, three-way Krippendorff $\alpha = 0.265$. Pairwise agreement separates leakage-prone and leakage-free graders cleanly: RadFact $\leftrightarrow$ VERT $\alpha = 0.701$; GREEN $\leftrightarrow$ RadFact $\alpha = 0.042$; GREEN $\leftrightarrow$ VERT $\alpha = -0.001$. The two MIMIC-free graders agree at the level of substantial radiologist inter-rater agreement in the Landis–Koch band, while the MIMIC-pretrained grader’s label distribution diverges from both MIMIC-free graders. This is direct empirical support for reporting silver-label agreement by subset rather than aggregating all three graders into a single statistic. The 2-of-3 majority rule assigns 1,599 supported, 335 contradicted, 773 uncertain, with 366 claims excluded for full three-way disagreement and 42 flagged

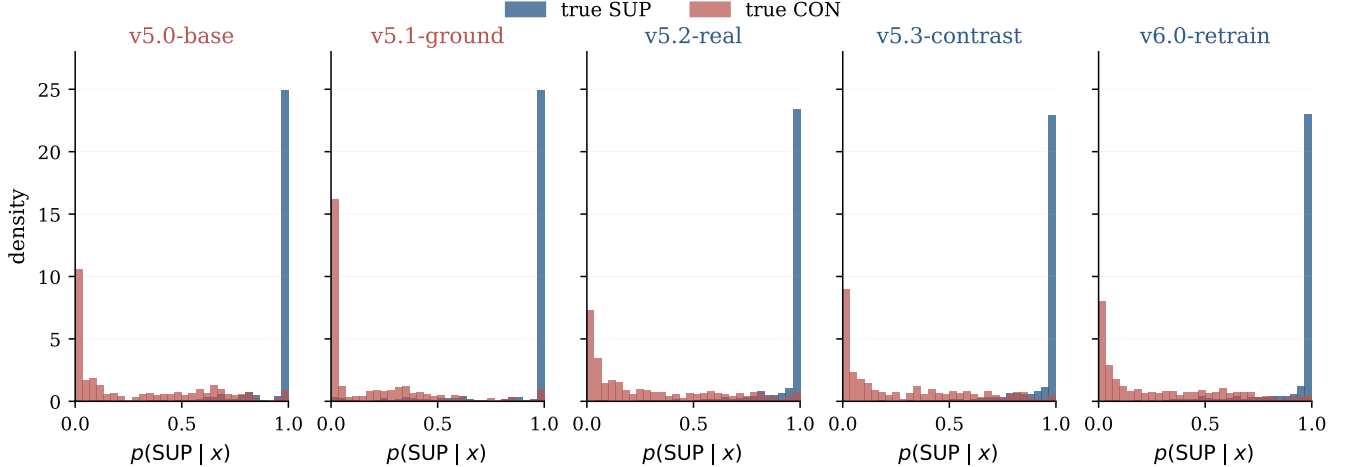


Figure 3: Conformal selection collapses on all but v5.3. Support-score distribution $p(\text{SUP} | x)$ for each trained configuration, on the 2,974-claim v6 test split, split by the true label. The two evidence-blind configurations (v5.0, v5.1) assign $p(\text{SUP} | x)$ values near the decision boundary to many contradicted rows, which is exactly the case that collapses inverted Benjamini–Hochberg selection to $n_{\text{green}} = 0$. v5.3-contrast and v6.0-retrain push contradicted rows to $p(\text{SUP} | x) \approx 0$ and supported rows to $p(\text{SUP} | x) \approx 1$, leaving a tail that BH can exploit.

low confidence.

5. Conclusion

We introduced a counterfactual diagnostic for multi-modal medical claim verification and used it to show that a cross-entropy baseline at 90.9% test accuracy is image-blind: zeroing the image drops accuracy by only 2.03 percentage points. Adding a grounding loss alone does not fix this. A consistency loss combined with an adversarial hypothesis-only filter closes the image-masking gap to 62.90pp, and contrastive evidence loss extends it to 69.24pp, both without degrading aggregate accuracy. A confirmatory retraining under the same loss composition on the v6 cohort reaches 70.21pp at 91.1% accuracy. External baselines fail the diagnostic on at least one axis, and leave-one-site-out training shows the mitigation does not survive distribution shift, making the same three metrics useful as a deployment-shift alarm.

Our contribution has three limitations. First, the benchmark is compact at approximately 7,500 images and 25,176 labeled training claims. PadChest-GR records were ingested into the cohort but were not normalized into (claim, evidence, label) training tuples in time for this submission; they currently play a silver-label validation role rather than a third training site, and the cross-site leave-one-out table reports two held-out configurations rather than three. Second, the training-time mitigation is training-distribution-specific. Under leave-one-site-out, the image-masking gap regresses to 3.21pp on held-out OpenI and the evidence-

shuffling gap drops to 2.69pp on held-out ChestX-Det10 — both below the evidence-blind threshold. The diagnostic flags this regression, but the mitigation does not prevent it. Third, the image-perturbation gap stays in the $[-0.88, +0.88]$ pp band across every configuration we trained. Horizontally flipping the image does not change the verdict even on laterality-turning claims. This is an architectural residual: the consistency loss penalizes the model for ignoring image presence but not for ignoring spatial arrangement. All training runs and diagnostic evaluations were executed on a single seed under compute-budget constraints; multi-seed replication is deferred to an extended version. A further caveat applies to the real-hallucination evaluation: we decompose the 1,500 generated reports into atomic claims via a Claude Haiku prompt and did not measure the faithfulness of that decomposition against a radiologist-split reference set; the silver-label ensemble α -statistics reported in Section 4 therefore assume the atomic-claim boundaries are themselves reliable.

Three directions follow from these limitations. The laterality residual requires an image-encoder-level intervention finer than the frozen BiomedCLIP patch resolution. The cross-site regression suggests that training-distribution mitigations should be paired with distribution-shift-aware methods such as weighted conformal inference [15] or multi-site curriculum training. The silver-label validation against PadChest-GR’s 11,995 radiologist-labeled sentences yielded 12 paired-image matches under a Jaccard-0.30 threshold; the remaining pairs fail because model-generated reports use

generic normal-pattern vocabulary while PadChest-GR sentences use specific pathology terms. A clinical-NER-based matcher that is finding-aware, rather than token-overlap-based, is the natural next step.

Counterfactual evidence-sensitivity testing should be a standard evaluation step before aggregate-accuracy reporting for any multimodal scientific model whose downstream use depends on its image or evidence inputs. The cost is a small number of additional forward passes, and the information it provides cannot be extracted from aggregate accuracy curves. Benchmark maintainers should report text-only and image-masked baselines alongside full-model results, as the NLI community did after partial-input baselines [7, 13] entered standard practice. Workshop and journal submission checklists in multimodal domains should require partial-input baselines in the same way reproducibility checklists now require dataset statistics. In safety-critical settings, any claim-level verifier positioned upstream of expert review should publish IMG and ESG numbers as part of its model card, so that deployment decisions are not made on aggregate accuracy alone.

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